In [1]:
```
pip install pandas numpy seaborn matplotlib scikit-learn
```

```
Requirement already satisfied: pandas in C:\Users\Abishek\anaconda3\Lib\site-packages (3.0.3)
Requirement already satisfied: numpy in C:\Users\Abishek\anaconda3\Lib\site-packages (2.4.6)
Requirement already satisfied: seaborn in C:\Users\Abishek\anaconda3\Lib\site-packages (0.13.2)
Requirement already satisfied: matplotlib in C:\Users\Abishek\anaconda3\Lib\site-packages (3.11.0)
Requirement already satisfied: scikit-learn in C:\Users\Abishek\anaconda3\Lib\site-packages (1.9.0)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\Abishek\anaconda3\Lib\site-packages (from pand
as) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\Abishek\anaconda3\Lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\Abishek\anaconda3\Lib\site-packages (from matplotlib
) (1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\Abishek\anaconda3\Lib\site-packages (from matplotlib) (0
.12.1)
Requirement already satisfied: fonttools>=4.22.0 in C:\Users\Abishek\anaconda3\Lib\site-packages (from matplotli
b) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\Abishek\anaconda3\Lib\site-packages (from matplotli
b) (1.4.9)
Requirement already satisfied: packaging>=20.0 in C:\Users\Abishek\anaconda3\Lib\site-packages (from matplotlib)
(26.0)
Requirement already satisfied: pillow>=9 in C:\Users\Abishek\anaconda3\Lib\site-packages (from matplotlib) (12.3
.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\Abishek\anaconda3\Lib\site-packages (from matplotlib) (3
.2.5)
Requirement already satisfied: scipy>=1.10.0 in C:\Users\Abishek\anaconda3\Lib\site-packages (from scikit-learn)
(1.18.0)
Requirement already satisfied: joblib>=1.4.0 in C:\Users\Abishek\anaconda3\Lib\site-packages (from scikit-learn)
(1.5.3)
Requirement already satisfied: narwhals>=2.0.1 in C:\Users\Abishek\anaconda3\Lib\site-packages (from scikit-lear
n) (2.23.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in C:\Users\Abishek\anaconda3\Lib\site-packages (from scikit
-learn) (3.5.0)
Requirement already satisfied: six>=1.5 in C:\Users\Abishek\anaconda3\Lib\site-packages (from python-dateutil>=2
.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib as mp
import sklearn as sk
```

In [5]:
```
df = pd.read_csv(r"C:\Users\Abishek\Downloads\house_prices_practice.csv")
```

In [6]:
```
df.head()
```

Out[6]:

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt	FullBath	BedroomAbvGr	LotArea	SalePrice
0	1	7	1560	0	1658	1969	2	1	8059	177106
1	2	4	2827	2	1319	2012	3	4	13530	301044
2	3	8	3920	0	841	2010	1	4	9010	360609
3	4	5	3044	0	1058	1998	0	4	13207	240556
4	5	7	801	1	2428	2020	0	1	9117	193656

In [7]:
```
df.tail(5)
```

Out[7]:

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt	FullBath	BedroomAbvGr	LotArea	SalePri
295	296	1	3495	1	1792	1954	2	5	4978	2506
296	297	5	3438	3	1266	2003	0	1	9373	3299
297	298	6	1992	0	1148	1996	1	1	7907	1846
298	299	3	3722	1	1407	1998	1	1	8097	3033
299	300	9	1667	0	988	1958	2	1	14553	2049

In [8]:
```
df.describe()
```

Out[8]:

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt	FullBath	BedroomAbvGr
count	300.000000	300.000000	300.000000	300.000000	300.000000	300.000000	300.000000	300.000000
mean	150.500000	5.326667	2307.386667	1.330000	1468.796667	1986.163333	1.523333	2.926667
std	86.746758	2.873001	1042.561303	1.109898	672.333705	21.377089	1.131543	1.456604
min	1.000000	1.000000	504.000000	0.000000	303.000000	1950.000000	0.000000	1.000000
25%	75.750000	3.000000	1392.250000	0.000000	903.000000	1967.000000	0.000000	2.000000
50%	150.500000	5.000000	2265.500000	1.000000	1502.000000	1986.000000	2.000000	3.000000
75%	225.250000	8.000000	3306.500000	2.000000	2129.500000	2004.250000	3.000000	4.000000
max	300.000000	10.000000	3998.000000	3.000000	2492.000000	2023.000000	3.000000	5.000000

In [9]:
```
df.shape
```

Out[9]:
```
(300, 10)
```

In [10]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Id            300 non-null    int64
 1   OverallQual   300 non-null    int64
 2   GrLivArea     300 non-null    int64
 3   GarageCars    300 non-null    int64
 4   TotalBsmtSF   300 non-null    int64
 5   YearBuilt     300 non-null    int64
 6   FullBath      300 non-null    int64
 7   BedroomAbvGr  300 non-null    int64
 8   LotArea       300 non-null    int64
 9   SalePrice     300 non-null    int64
dtypes: int64(10)
memory usage: 23.6 KB
```

In [11]:
```
df.nunique().sum()
```

Out[11]:
```
np.int64(1557)
```

In [12]:
```
df.head(10)
```

Out[12]:

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt	FullBath	BedroomAbvGr	LotArea	SalePrice
0	1	7	1560	0	1658	1969	2	1	8059	177106
1	2	4	2827	2	1319	2012	3	4	13530	301044
2	3	8	3920	0	841	2010	1	4	9010	360609
3	4	5	3044	0	1058	1998	0	4	13207	240556
4	5	7	801	1	2428	2020	0	1	9117	193656
5	6	10	680	1	2480	1950	2	1	13081	213952
6	7	3	1106	2	1096	1962	1	4	11153	145539
7	8	7	3670	1	437	2000	2	5	4989	350830
8	9	8	1199	2	1653	2005	1	5	14040	275955
9	10	5	1492	1	2175	2011	1	2	11218	211664

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
Id             0
OverallQual    0
GrLivArea      0
GarageCars     0
TotalBsmtSF    0
YearBuilt      0
FullBath       0
BedroomAbvGr   0
LotArea        0
SalePrice      0
dtype: int64
```

In [14]:
```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

In [15]:
```
slr = LinearRegression()
```

In [16]:
```
x = df[['Id', 'OverallQual', 'GrLivArea', 'GarageCars', 'TotalBsmtSF',
    'YearBuilt', 'FullBath', 'BedroomAbvGr', 'LotArea']]

y = df['SalePrice']
```

In [17]:
```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.20, train_size=0.80, random_state=12
)
```

In [18]:
```
slr.fit(x_train, y_train)
pred = slr.predict(x_test)
print("R2 Score:", slr.score(x_test, y_test))
```

```
R2 Score: 0.9776119778020329
```

In [ ]: